DNA migration using flashing ratchet potential

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Abstract

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1 Introduction

The separation of microscopic particles according to their physical properties is a central problem in fields such as biophysics, chemistry, and materials science. In particular, the ability to sort biological macromolecules such as DNA by size is essential for analysis and characterization of complex samples. Traditional techniques, such as centrifugation, rely on external forces and macroscopic gradients, but alternative methods based on stochastic transport have attracted significant interest.

One such approach exploits Brownian ratchets, where thermal fluctuations are rectified into directed motion using an asymmetric, time-dependent potential. Although the motion of particles in solution is inherently random due to thermal agitation, it is possible to bias this motion by periodically switching ("flashing") an asymmetric potential. This leads to a net drift, even in the absence of a net external force, a phenomenon often referred to as a Brownian motor.

In this work, we investigate a model system consisting of particles undergoing Brownian motion in a one-dimensional flashing ratchet potential. The aim is to study how the resulting drift velocity depends on particle size and on the flashing period of the potential, and to assess the feasibility of using this mechanism for particle separation. The problem is approached numerically using a stochastic differential equation framework based on the Langevin equation.

2 Experimental Method

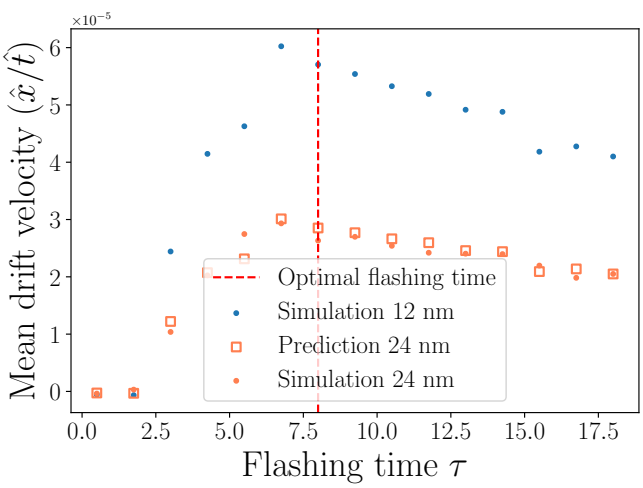

Figure 1: Drift velocity as a of flashing time.

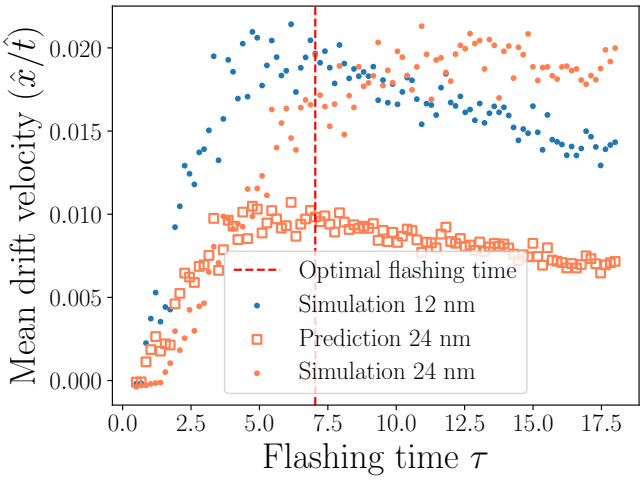

Figure 2: Drift velocity as a function of flashing time.

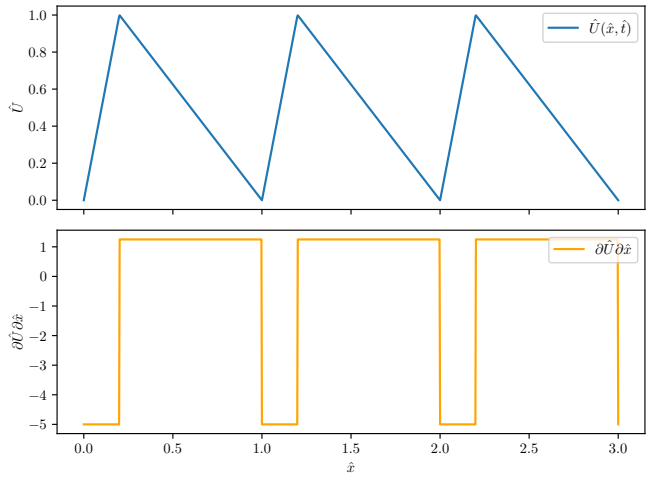

Figure 4: Plotted shape of potential and force

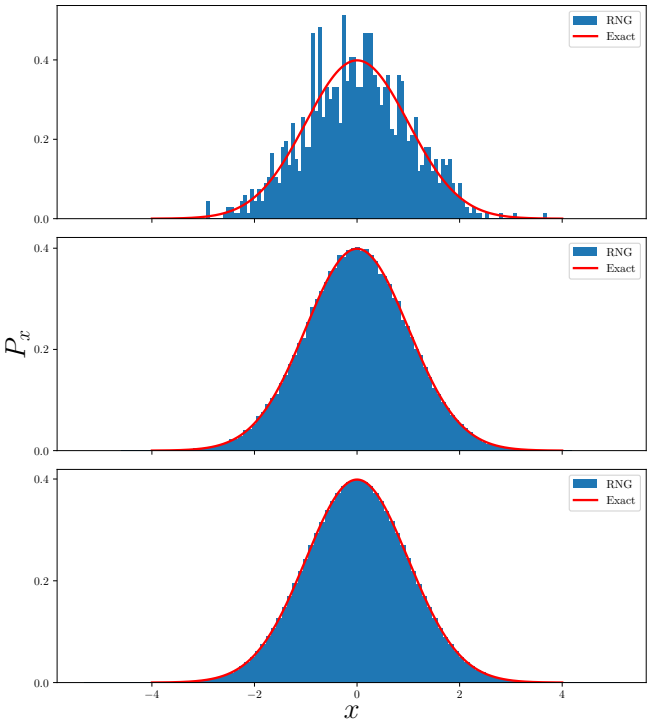

Figure 3: Histogram of Gaussian random numbers generated by the Box-Muller algorithm. The red curve represents the probability density function of a Gaussian distribution with the same mean and standard deviation as the generated samples.

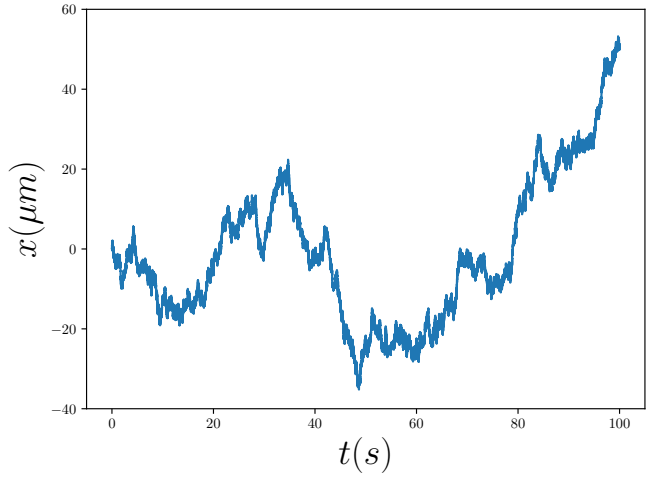

Figure 5: Plotted shape of potential and force $\Delta U = 0.1k_BT$

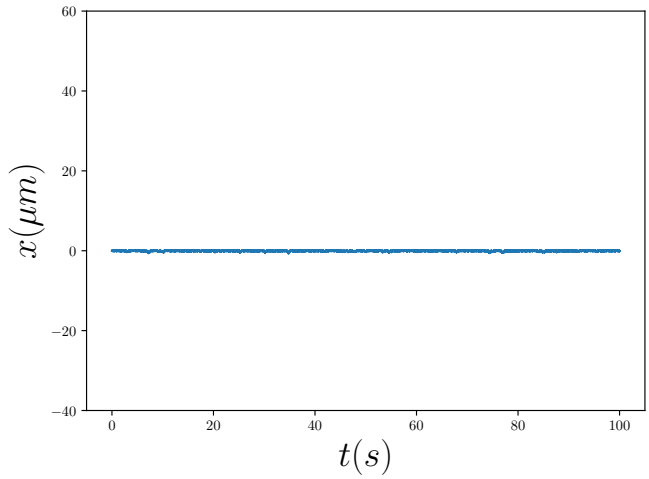

Figure 6: Plotted shape of potential and force $\Delta U = 10k_BT$

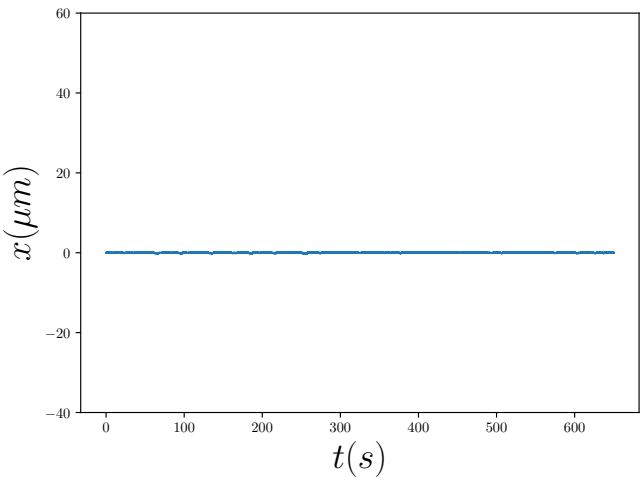

Figure 7: Traveled path at $\Delta U = 80eV$

$$U(x,t) = U_r(x)f(t), \tag{1}$$

$$U_r(x) = \begin{cases} \frac{x}{\alpha}\Delta U, & 0 \leq x < \alpha L \\ \frac{L-x}{\alpha-1}\Delta U, & \alpha L \leq x < L \end{cases} \tag{2}$$

$$f(t) = \begin{cases} 1, & t < t_0 \\ 0, & t \geq t_0 \end{cases} \tag{3}$$

$$x_{n+1} = x_n - \frac{1}{\gamma_i}\frac{\partial U}{\partial x}(x_n, t_n)\delta t + \sqrt{\frac{2k_B T \delta t}{\gamma_i}}\hat{\xi}_n, \tag{4}$$

$$|x_n - x| << \alpha L \tag{5}$$

Choosing δt such that it the equation

$$\frac{1}{\gamma_i}\max\left|\frac{\partial U}{\partial x}\right|\delta t << \alpha L \tag{6}$$

is satisfied, ensures that the particle does not move more than a fraction of the potential's spatial period in one time step. This is crucial for accurately capturing the dynamics of the system, as it allows for a more precise representation of the particle's motion within the potential landscape. By keeping the time step sufficiently small, we can ensure that the numerical integration of the particle's trajectory remains stable and accurate, preventing any significant overshooting that could lead to erroneous results in our simulations.

2.1 Reduced units

To simplify our calculations and reduce the number of parameters in our system, we can introduce reduced units. This involves choosing characteristic scales for length, energy, and time, which allows us to express all quantities in terms of these scales.

$$\hat{x}_n = \hat{x} \tag{7}$$

$$\hat{x} = \frac{x}{L}, \quad \hat{t} = \omega t, \quad \omega = \frac{\Delta U}{\gamma_i L^2}, \tag{8}$$

$$\hat{U}(\hat{x}, \hat{t}) = \frac{U(x,t)}{\Delta U}, \quad \hat{D} = \frac{k_B T}{\Delta U} \tag{9}$$

$$p(U) = \frac{e^{-\frac{U}{k_B T}}}{k_B T(1 - e^{-\frac{\Delta U}{k_B T}})} \tag{10}$$

$$n(x,t) = \frac{N}{\sqrt{4\pi D t}}\exp\left(-\frac{x^2}{4Dt}\right) \tag{11}$$

$$m\frac{d^2\mathbf{x}}{dt^2} = -\nabla U(\mathbf{x}, \mathbf{t}) - \gamma\frac{\mathbf{dx}}{\mathbf{dt}} + \xi(\mathbf{t}) \tag{12}$$

3 Results

4 Discussion

5 Conclusion

6 Acknowledgements

7 Author Contributions

I coordinated with **De jeg jobbet med** to review and validate results and methods during the project.

8 AI declaration

References